

Quiz 3 [15 points]

Pointer Arithmetic & Memory Understanding

1 Pointer Arithmetic [10 points]

Implement a function called `copy_reverse` that copies the contents of one integer array into another array in **reverse order**, using only **pointer arithmetic** (no array indexing or functions from `string.h`).

```
#include <stdio.h>

void copy_reverse(int *dest, int *src, int n);

int main(void) {
    int nums[] = {1, 2, 3, 4, 5};
    int reversed[5];

    copy_reverse(reversed, nums, 5);

    for (int i = 0; i < 5; i++)
        printf("%d ", reversed[i]);
    printf("\n");    // Output should be: 5 4 3 2 1

    return 0;
}

void copy_reverse(int *dest, int *src, int n) {
    /* Your code here (use only pointer arithmetic) */
}
```

2 Pointer Knowledge [5 points]

Answer the following three questions briefly.

2.1 [2 points]

Given an array `int arr[10];`, give two valid expressions to access the **address** of the `i`th element.

2.2 [2 points]

What is wrong with this code?

```
int* make_ptr(void) {  
    int x = 42;  
    return &x;  
}  
  
int main(void) {  
    int *p = make_ptr();  
    printf("%d\n", *p);  
}
```

2.3 [1 point]

What happens when you dereference a NULL pointer?

End of Quiz